Curriculum Vitae

Dr. Parth **Nayak**

Data Scientist | AI Researcher | Astrophysicist

- Munich, Germany
- par-nay.github.io
- parth.nayak@physik.lmu.de
- dr-parth-nayak-astro
- par-nay

Education

- **Ph.D. in Astrophysics** (Dr. rer. nat.)
  LMU Munich | Dec 2025
- **B.Sc.+M.Sc. in Physics** (dual degree)
  IISER Bhopal | Jul 2020

Skills

ML / Deep Learning

Neural networks (CNNs, MLPs), simulation-based inference (SBI), generative models (normalizing flows, diffusion models, autoencoders, GANs)

Other AI

LLM prompt engineering, evolutionary algorithms, Gaussian processes, Bayesian & Monte Carlo methods, mixture models, transfer learning

Programming

Python(proficient); HTML, CSS, Javascript (intermediate); C++, Java (basic)

Python packages

TensorFlow, Keras, Optuna, NumPy, SciPy, Pandas, Matplotlib, H5Py, AstroPy, HealPy, *among others*

Other IT

LaTeX typesetting, High-performance computing (HPC), Git version control, Markdown, Mac/MS Office, Shell (bash, Z shell)

Languages

English (full working proficiency); Gujarati, Hindi (native/bilingual); German (B1/intermediate); French (beginner)

Highlights

- **5+ years** of **research experience** in **Data Science** and **AI** algorithm development
- **Authored 4 research papers** on novel AI applications in astrophysics
- Delivered **10+ technical talks** at **international conferences** and **research institutes**
- Received **5+ academic awards** and **fellowships**

Relevant experience

Doctoral Candidate: Astrophysics & Data Science *LMU Munich | Oct 2021 - Dec 2025*

Hightech Agenda Bayern chair of Astrophysics, Cosmology & AI + Munich Center for Machine Learning (MCML)

- **Developed a ML inference framework "*LyaNNA*"** [par-nay/lyanna] based on a novel 1D ResNet with Python/TensorFlow from scratch that **outperforms conventional methods** by **>100%** in inference precision for complex, realistic, noisy spectroscopic data (simulation-based inference)
- **Developed "*SynTH*"** [par-nay/synth] to model intergalactic absorption from massive cosmological **hydrodynamic simulations** and **synthesize training data** for *LyaNNA*
- **Authored 3** research papers on AI for inference in astrophysics [Nayak+ 2024, 2025, Chang+ 2025]

Master's Thesis: Astrophysics & Data Science *IISER Bhopal | Aug 2019 - Jul 2020*

- **Developed a Genetic Algorithm framework "*GA-ILC*"** to optimize the reconstruction of cosmic microwave background (CMB) from contaminated observations
  - A novel, **scalable** evolutionary algorithm [par-nay/geneticalpy] to extract **faint cosmological signals** from **high-noise** environments (foreground removal)
- **Authored one** research paper on evolutionary algorithms in astrophysics [Nayak & Saha 2022]

Leadership & supervision

- **Co-supervisor of LMU Master's theses and labs** *Oct 2021 - Dec 2025*
  Designed and supervised **two LMU Master's theses**, and co-supervised **5+** computational and instrumental **lab courses**
- **Co-organizer of "Astronomy on Tap" (AoT) Munich** *since Oct 2025*
  Coordinating the **monthly outreach event** to engage the general **public** with accessible scientific talks and pub-quizzes
- **Co-organizer of "Meet the Speaker" @LMU Observatory** *Oct 2023 - May 2025*
  Revitalized the weekly series by relocating it to the historic 200-year old telescope dome, a unique atmosphere for students for informal **networking** with **international experts**
- **Coordinator & Senior Advisor of IISERB Astronomy Club (IBAC)** *2016 - 2020*
  Played a foundational role as the club's second coordinator, establishing enduring traditions and **growing the community** into one of India's most popular student astronomy clubs
- **Physics Department Representative** *Aug 2017 - Jul 2018*
  Unanimously elected to serve as the **primary liaison** between faculty and a class of **60+** physics major students

Selected awards & fellowships

- **DAAD Research Grant — Ph.D. scholarship** *Germany | Oct 2021 - Dec 2025*
- **Proficiency Medal in Physics — IISER Bhopal** *India | Jul 2020*
- **DAAD-WISE Scholarship** *Germany | May - Jul 2019*
- **INSPIRE Scholarship for Higher Education** *India | Aug 2015 - Jul 2020*